

• Assignment No: 2 •

Measures of Central Tendency

Que 1] Calculate the A.M for the following data of weights in kgs.

Weights (in kgs): 58, 55, 75, 45, 61, 62, 75, 45, 47, 51, 55, 52.

$$\bar{X} = \frac{\sum X}{N} = \frac{58+55+75+45+61+62+75+45+47+51+55+52}{12}$$

$$= \frac{681}{12} = 56.75 \text{ kgs.}$$

$$\therefore \bar{X} = 56.75 \text{ kgs}$$

Que 2] Calculate A.M from the data given below.

X: 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30.

f: 2, 8, 10, 12, 5, 17, 11, 9, 5, 4, 1

X	f	X.f
10	2	20
12	8	96
14	10	140
16	12	192
18	5	90
20	17	340
22	11	242
24	9	216
26	5	130
28	4	112
30	1	30
N = 84		$\sum X.f = 1608$

$$\bar{X} = \frac{\sum X.f}{N}$$

$$= \frac{1608}{84}$$

$$= \underline{\underline{19.1429}}$$

Q.3] Find A.M income from the data of 53 employees.

Income (in '00 Rs):	100-200	200-300	300-400	400-500	500-600
No. of employees	2	8	9	10	13
	600-700	700-800			
	7	4			

Income (x)	f (No. of employees)	Midpoint	f.m
100-200	2	100	200
200-300	8	250	2000
300-400	9	350	3150
400-500	10	450	4500
500-600	13	550	7150
600-700	7	650	4550
700-800	4	750	3000
	$\Sigma f = N = 53$		$\Sigma f.m = 24550$

$$\bar{X} = \frac{\Sigma f.m}{N} = \frac{24550}{53} = \frac{24550}{53} = \underline{\underline{463.2075}}$$

Q.4] Find the Median from the following data.

Marks	0-10	10-20	20-30	30-40	40-50
No. of Students	4	14	22	8	2

Marks (x)	No. of students (f)	Cf
0-10	4	4
10-20	14	18
20-30	22	40
30-40	8	48
40-50	2	<u>50</u>
$\Sigma f = 50$		

Here

Median = size of $\left(\frac{N+1}{2}\right)^{\text{th}}$ item in data.

= size of $\left(\frac{50+1}{2}\right)^{\text{th}}$ item in data

= size of 25.5^{th} item in data

To see the C.f. which is equal to or just greater than 25.5 is 40.

$\therefore$ The corresponding class is 20-30.

$\therefore$ Median class is 20-30.

$$\therefore \text{Median} = L_1 + \left(\frac{\frac{N}{2} - fcf}{f} \right) \times i$$

$$= 20 + \left(\frac{\frac{50}{2} - 18}{22} \right) \times 10$$

$$\therefore \text{Median} = \underline{\underline{23.18}}$$

Q.5] Calculate Median for the following Data.

Marks.	0-10	10-20	20-30	30-40	40-50	50-60	60-70
No. of students	5	7	8	12	28	22	10

70-80

8

Marks (X)	No. of Students	c.f
0-10	5	5
10-20	7	12
20-30	8	20
30-40	12	32
40-50	28	60
50-60	22	82
60-70	10	92
70-80	8	100
$\Sigma f = N = 100$		

Here

Median = Size of $\left(\frac{N+1}{2}\right)^{\text{th}}$ item in data

= size of $\left(\frac{100+1}{2}\right)^{\text{th}}$ item in data.

= size of 50.5th item in data.

To see the c.f which is equal to or just greater than 50.5 is 60.

∴ The corresponding class is 40-50.

∴ Median class is 40-50.

$$\text{Median} = L_1 + \frac{\left(\frac{N}{2} - f.c.f\right)}{f} \times i$$

$$= 40 + \frac{\left(\frac{100}{2} - 32\right)}{28} \times 10$$

$$\boxed{\text{Median} = 46.43}$$

Q.6] Calculate Mode for the following Data.

class	10-20	20-30	30-40	40-50	50-60	60-70	70-80
f	24	42	56	66	108	130	154

Here the Maximum frequency is 154
Therefore Corresponding Mode class is 70-80

Formula:

$$\text{Mode} = L_1 + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i$$

$$= 70 + \frac{154 - 130}{2 \times 154 - 130 - 0} \times 10$$

$$\text{Mode} = 71.35$$

Q.7] Calculate Mean, Median, Mode for the following data.

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No of stu	4	2	18	22	21	19	10	3	1

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Marks (X)	frequency f	midpoint	f.m	cf
0-10	4	5	20	4
10-20	2	15	30	6
20-30	18	25	450	24
30-40	22	35	770	46
40-50	21	45	945	67
50-60	19	55	1045	86
60-70	10	65	650	96
70-80	3	75	225	99
80-90	1	85	85	100
	$N=100$	Σmi	$\Sigma = 4320$	

1] Mean

$$\therefore \text{Mean}(\bar{x}) = \frac{\Sigma fmi}{N} = \frac{4320}{100} = 43.20$$

2] Median:-

Here,

Median = size of $\left(\frac{N+1}{2}\right)^{\text{th}}$ item in data

= size of $\left(\frac{100+1}{2}\right)^{\text{th}}$ item in data

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= size of 50.5^{th} item in the data

To see the c.f which is equal to or just greater than 50.5 is 67.

$\therefore$ The Corresponding class is 40-50.

$\therefore$ Median class is 40-50.

$$\begin{aligned}\therefore \text{Median} &= L_1 + \frac{\left(\frac{N}{2} - f.c.f\right)}{f} \times i \\ &= 40 + \frac{\left(\frac{100}{2} - 46\right)}{22} \times 10\end{aligned}$$

$$\text{Median} = 41.90$$

3] Mode:-

Here the Maximum frequency is 22.
Therefore the Mode class is 30-40.

$$\therefore \text{Mode} = L_1 + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i$$

$$= 30 + \frac{22 - 18}{2(22) - 18 - 21} \times 10$$

$$= 30 + \frac{4}{5} \times 10$$

$$\text{Mode} = 38$$

Q.8] Find the Mean, Median, Mode for the following data.

Wages in Rs.	30-40	40-50	50-60	60-70	70-80.
No. of Workers.	9	13	25	11	7.

x_i	f_i	m_i	$f_i m_i$	cf
30-40	9	35	315	9
40-50	13	45	585	22
50-60	25	55	1375	47
60-70	11	65	715	58
70-80	7	75	525	65
	$\Sigma f_i = 65$		$\Sigma = 3515$	

1] Mean

$$\bar{X} = \frac{\Sigma f_i m_i}{N} = \frac{3515}{65} = 54.07$$

2] Median.

Here,

$Me = \text{size of } \left(\frac{N+1}{2}\right)^{\text{th}} \text{ item in the data}$

$= \text{size of } \left(\frac{65+1}{2}\right)^{\text{th}} \text{ item in the data}$

$= \text{size of } 33^{\text{th}} \text{ item in the data.}$

To see c.f which is equal to or just greater than 33 is 47

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∴ The corresponding class is 50-60.
 ∴ The Median class is 50-60.

$$\begin{aligned}\therefore \text{Median} &= L + \left(\frac{\frac{N}{2} - \text{cf} \cdot f}{f} \right) \times i \\ &= 50 + \left(\frac{\frac{65}{2} - 22}{25} \right) \times 10\end{aligned}$$

$$\text{Median} = 54.20$$

3] Mode:-

∴ Here the Maximum frequency is 25
 Therefore the Mode class is 50-60.

$$\begin{aligned}\therefore \text{Mode} &= L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i \\ &= 50 + \frac{25 - 13}{2(25) - 13 - 11} \times 10 \\ &= 50 + \frac{12}{26} \times 10\end{aligned}$$

$$\text{Mode} = 54.61$$

Q.9] Find the Mean, Median, Mode for the following data.

Weight (kg)	30-40	40-50	50-60	60-70	70-80	80-90
Frequency	18	37	45	27	15	8

class	f_i	m_i	$f_i m_i$	C.f
30-40	18	35	630	18
40-50	37	45	1665	55
50-60	45	55	2475	100
60-70	27	65	1755	127
70-80	15	75	1125	142
80-90	8	85	680	150
	$\Sigma = 150$		$\Sigma = 8330$	

1] Mean =

$$\bar{X} = \frac{\Sigma f_i m_i}{N} = \frac{8330}{150} = 55.53$$

2] Median:-

Me = size of $\left(\frac{N+1}{2}\right)^{\text{th}}$ size item in the data

= size of $\left(\frac{150+1}{2}\right)^{\text{th}}$ item in the data

= size of 75.5^{th} item in the data

To see the C.f which is equal to or just greater than 75.5 is 100.

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The corresponding class is 50-60.

$$\therefore Me = L_1 + \frac{\left(\frac{N}{2} - \text{c.f.}\right)}{f} \times i$$

$$= 50 + \frac{\frac{150}{2} - 55}{45} \times 10$$

$$\text{Median} = 54.44$$

3] Mode

Here the High frequency is 45

$\therefore$ The Mode class is 50-60.

$$\therefore \text{Mode} = L_1 + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i$$

$$= 50 + \frac{45 - 37}{2(45) - 37 - 27} \times 10$$

$$= 50 + \frac{8}{26} \times 10$$

$$\text{Mode} = 53.07$$

Q.10] To calculate Mean, Median, Mode for the following data.

Score	0-5	5-10	10-15	15-20	20-25	25-30	35-40	40-45	45-50.
Frequency	6	12	50	120	225	250	185	110	32.

class	f_i	m_i	$m_i f_i$	c.f
0-5	6	2.5	15	6
5-10	12	7.5	90	18
10-15	50	12.5	625	68
15-20	120	17.5	2100	188
20-25	225	22.5	5062	413
25-30	250	27.5	6875	663
30-35	185	32.5	6012	848
35-40	110	37.5	4125	958
40-45	32	42.5	1360	990
45-50	10	47.5	475	1000
	$N=1000.$		$\Sigma=26739$	

1] Mean =
$$\bar{x} = \frac{\Sigma m_i f_i}{N} = \frac{26739}{1000} = 26.73$$

2] Median:-

Here,

Median = size of $\left(\frac{N+1}{2}\right)^{\text{th}}$ item of the data

= size of $\left(\frac{1000+1}{2}\right)$ item in the data.

= size of 500.5^{th} item in the data.

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To see the C.F which is equal to or just greater than 500.5 is 663.

∴ The Median class is 25-30.

$$\therefore \text{Median} = L_1 + \left(\frac{\frac{N}{2} - \text{Pcf}}{f} \right) \times i$$

$$= 25 + \left(\frac{\frac{1000}{2} - 413}{250} \right) \times 5$$

$$\therefore \text{Median} = 26.74$$

3] Mode

Here the High frequency is 250.

∴ Therefore the Mode class is 25-30.

$$\therefore \text{Mode} = L_1 + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i$$

$$= 25 + \frac{250 - 225}{2(250) - 225 - 185} \times 5$$

$$= 25 + \frac{25}{90} \times 5$$

$$\text{Mode} = 26.3889$$